

Scent as a Sensory Modality for Data Physicalisation for Office Well-being

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ABSTRACT

While the office environment is increasingly viewed as an opportune setting to improve the health and well-being of individuals, the use of scent to physicalize data in this field is mostly under-explored. To study the role of olfactory artifacts in the context of office well-being, we developed two artifacts: Re-cafeen and Whiff. Both artifacts use different scent modalities based on different data types and spatial placements. The used scents in the artifacts are based on a study of 10 scents which asked participants (N=13) to express which scents correspond to the following activities: (i) having an active meeting, (ii) taking a break from work, and (iii) working behind a standing desk. The artifacts were evaluated using smell space mapping. Based on this evaluation, we reflect on the challenges for future work on this topic. Our insights are relevant for the development of olfactory artifacts on the topic of office well-being.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Olfactory Design, Office Well-being, Data physicalisation, Scent

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1 INTRODUCTION

In recent years, the office environment has been explored as an opportune setting to improve the health and well-being of individuals [12, 29]. The office environment is one of the main settings

where individuals display inactive and sedentary behavior due to work that involves prolonged sitting, leading to an increased risk of having a non-active and/or sedentary lifestyle, associated with non-communicable diseases such as type II diabetes and cardiovascular diseases [5, 24, 27, 57]. The quality of the office environment itself – such as ventilation, temperature, but also office workers' posture and activity pattern – show a significant impact on the work performance, and overall office well-being of office workers [2, 21, 39, 52]. To address these issues on office well-being, prior work explored tangible artifacts to monitor, communicate, share and, improve the health of office workers [12]. Such artifacts aim to, for example, promote active ways of working [13, 14], visualize the productivity of individuals to indicate when to take a break [9, 62] or use sensing to indicate whether or not the office environment of individuals is a healthy office setting [8, 11, 37, 62]. These artifacts use several forms of physicalisation to present and visualize their feedback in the work environment, which shows a promising route for the communication of office well-being.

There is an encouragement in the Human-Computer Interaction (HCI) community to use olfactory output and to explore the properties of scent thorough design due to the peripheral qualities of scent, making it ideally suited as an ambient and calm displays [33, 42, 48–50]. The use of scent to physicalize data in the design space of office well-being is under-explored, as it remains focused on the more traditional interactions that trigger the senses of sight, hearing, and touch [12]. The use of scent is used as a means to warn and activate individuals in certain situations but also triggers emotions and memories from former times [49]. Scent is different in the way it is directly linked to the Limbic system, which is the part of the brain that deals with memories and emotions [15, 56, 61]. The emotional reaction of individuals towards a scent is determined by association, causing certain smells to be perceived differently from person to person. Additionally, several studies have been conducted to learn how scent stimuli can reduce stress [3, 65] or help with multitasking [28, 30, 43]. Scent could also be used as a form of information decoration, where individuals to whom the information is less or not relevant, can still benefit from the presentation/visualization of data [18]. This is especially seen in the emerging field of olfactory-focused design which shows promising areas where scent is used as an interaction modality in design [42].

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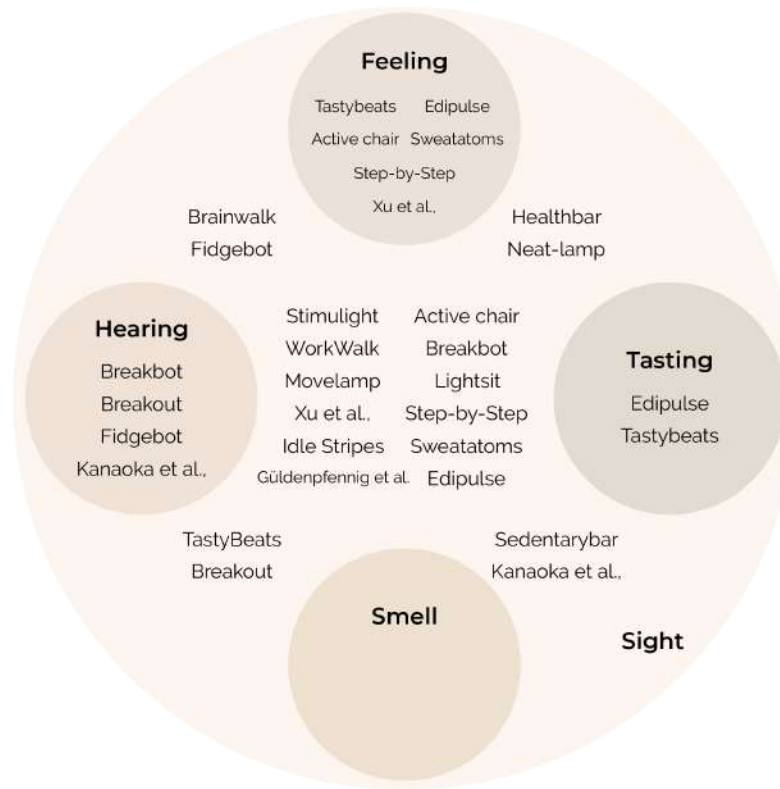


Figure 1: Overview of the tangible artifacts, indicated by artifact name or author (in case artifact did not have a name) developed in the field of office well-being. All interventions are divided based on the senses they trigger with hearing [6, 20, 32, 53], taste [34, 36], feeling [34–36, 51, 54, 66], sight [1, 6, 7, 14, 20, 22, 25, 26, 32, 34–36, 44, 46, 51, 53–55, 64, 66] and smell.

To study the role of olfactory artifact in the context of office well-being, we developed two artifacts called Re-cafeen and Whiff. Both artifacts use different scent modalities based on different data types and spatial placements to create healthier work environments. The implemented scents in the artifacts are selected, based on a study that asked participants ($N = 13$) to reflect on a selection of 10 scents and express which scents correspond to the following activities: (i) having an active/non-sedentary meeting (e.g., a WorkWalk [14]), (ii) taking a break from work and (iii) working behind a standing desk. The artifacts were evaluated using the Smell space mapping of Maggioni et al., [42]. Based on this evaluation, we reflect on the challenges for future work on this topic. The produced knowledge is relevant for the use and development olfactory focused artifacts on the timely topic of office well-being.

2 RELATED WORK

Several interventions are developed over the last decades to improve the well-being of office workers. To analyse this work, and their use of triggering senses in interventions, the review of Damen et al., [12] was analysed by two experts. The two experts in this field analysed the tangible interventions (total of 20 interventions) from this review. The purely digital interventions were not included

because these could only trigger two senses, namely sight, and hearing. The results of the analysis (Figure 1) show that all interventions ($20/20 f = 100\%$) trigger the sight of individuals. Next to triggering sight, interventions trigger the touch of individuals ($6/20 f = 30\%$). In lesser extent, taste ($2/20 f = 10\%$) and hearing ($4/20 f = 20\%$) are triggered. The smell of individuals in this field is not triggered by any of the interventions, showing an opportunity for this field.

Using scent as a medium to physicalize data for the well-being of office workers, comes with some challenges. The scent needs to be spread over (parts) of the office environment. It takes time to deliver and spread a scent over an environment and, therefore, for the user to notify [16]. Examples of ways to spread scent are the use of vaporization, airflow, heating, and atomization [67]. Vaporization is the process where a chemical substance is exposed and released in the air over time. This process can be accelerated by using airflow. Another way to spread the scent is via heat. Heat has the advantage that very small vessels with scents can be used and controlled and emitted almost instantly. Finally, atomization is used where a diffuser, sprayer, or ultrasonic waves are used to emit a mist of smell. However, this process has the disadvantage that it vaporizes over time, therefore only having a temporary effect [67].

An example of an artifact that implements scent in their design is SensaBubble [58]. SensaBubble is a chrono-sensory mid-air display

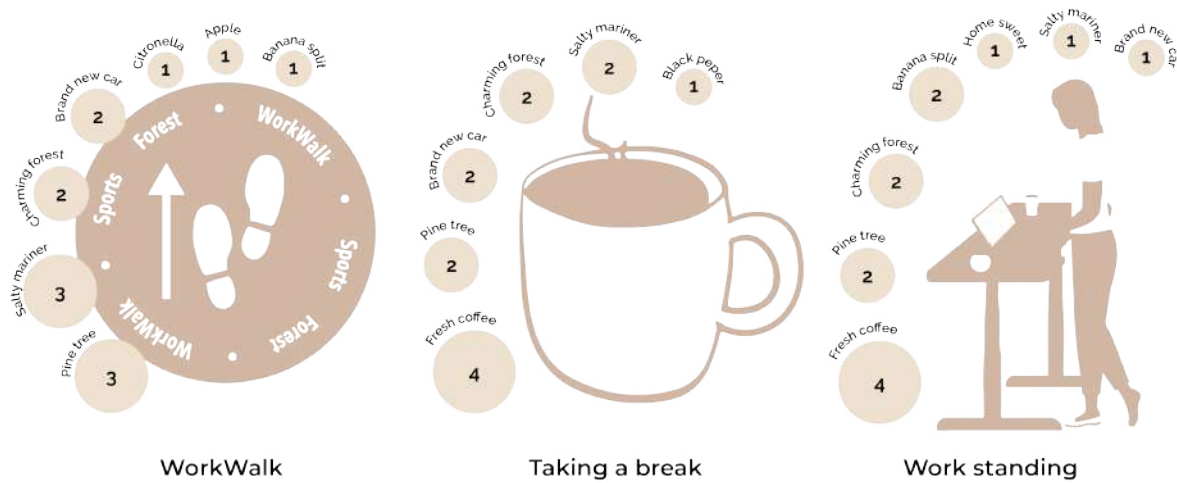


Figure 2: Overview of the results of the scent study. Numbers in the image represent how often a person indicated a smell as representative of a healthy way of working.

that generates scented bubbles to deliver notification and information to the user via a number of sensory modalities. The bubbles are filled with a scented fog allowing for the containment of a scent until the bubble bursts [58]. Next, inScent [16], is developed as a wearable olfactory display that is worn in everyday situations and allows the user to receive personal scented notifications. Users can individually use and assign scents to applications and use cases using modular scent cartridges based on their preferences. Smicons [33] also uses scent as a tool to send information. Ambient smell is used to give individuals all kinds of information including a personal reminder, stock updates or to provide awareness and connectedness. We build upon these examples by using them as inspiration on how to use scent as a sensory modality to enhance the work well-being of office workers.

3 METHOD: DESIGNING WITH SCENTS

Scents have a strong connection to the memories and emotions of individuals [61]. Scent can therefore give a more intense experience within an intervention, but only if the scent is related to the product (e.g., the research artifact) [41]. Having a scent which is conflicting with the corresponding visual information could have a negative effect on the usability of the intervention. It is therefore important that the scent which is spread is corresponding to the trigger that the intervention wants to evoke (e.g., having a healthier office workstyle).

3.1 Exploring different Scents

To find the corresponding activities with the correct scent, a study was conducted to learn if and how certain smells correspond to healthy ways of working. Participants ($N = 13$, 7 male, 6 female, age between 20 - 24) were asked to indicate which of a selection of 10 scents correspond to the following activities: (i) having an active/non-sedentary meeting (e.g., a WorkWalk [14]), (ii) taking a break from work and (iii) working behind a standing desk. These 3

activities were chosen due to them being to most common and/or promising ways of promoting healthy and active ways of working [12]. The 10 scents were selected based on a larger group of 15 scents from which scents with similar characteristics were removed. The 10 scents were titled: Banana split, Charming forest, Home sweet home, Fresh coffee, Black peppercorn, Brand new car, Apple, Salty mariner, Pine tree, and Citronella. The following protocol was followed after consent was obtained from the participants: *The user test involves 10 different smells which we will let you smell by using scented oils on perfume paper. The scents are labeled with numbers. Please pick each number up from the table when we ask you to. Additionally, we will ask questions that are the same for every scent.*

The results of the study (Figure 2) show that, when asked which scent corresponds mostly with having an active meeting (e.g., a WorkWalk), participants indicated scents that are related to nature and going outside. The scents Pine tree ($3/13 f = 23,1\%$), Salty mariner ($3/13 f = 23,1\%$), and Charming forest ($2/13 f = 15,4\%$) were all indicated as corresponding scents. The scent Brand new car was also corresponding to this activity ($3/13 f = 23,1\%$). For scents corresponding to taking a break, the scent of Fresh coffee ($4/13 f = 30,8\%$) was indicated most frequently. Similar to active meetings, nature-related scents were indicated with Pine tree ($2/13 f = 15,4\%$), Salty mariner ($2/13 f = 15,4\%$), and Charming forest ($2/13 f = 15,4\%$). The scent corresponding to working behind a standing desk was also the scent of fresh coffee ($4/13 f = 30,8\%$). The scents Banana split ($2/13 f = 15,4\%$), Charming forest ($2/13 f = 15,4\%$), and Pine tree ($2/13 f = 15,4\%$) were also indicated as corresponding.

The insights of the study were implemented into two artifacts. These artifacts called Whiff and Re-cafeen use different scents and mechanisms to use scent as a trigger to increase the well-being of office workers.



Figure 3: Whiff: an olfactory artifact that spreads a forest scent to create awareness on the posture of office workers.

4 ARTIFACTS

Based on the presented insights we developed two artifacts that implemented the studied mechanisms and scents. First, we discuss Whiff which uses the process of atomization to spread a forest-like scent to remind office workers to take a break and go for a WorkWalk [14]. Second, we present Re-cafeen which uses the process of heating to spread the scent of coffee to remind individuals to take a break from their heated meeting room. To evaluate the artifacts, the smell space mapping of Maggioni et al., [42] was used. The smell space mapping evaluates artifacts on four design features: (i) chemical, (ii) emotional, (iii) spatial, and (iv) temporal on their use of olfactory in the design space. The evaluation of the artifact was done based on the experience of the authors with the artifacts and the initial usability experiments.

4.1 Whiff

Whiff (Figure 3) was developed to make individuals reflect on their sitting time and posture. The device consists of two parts: a sensor system to measure the pressure and therefore the sitting position and a flex sensor, which is attached to the user's back to monitor the position of the back, and an output device, called Whiff, which notifies the user about their behaviour using scent. The two parts of the artifacts were controlled via a Particle Photon to make sure the two systems could work wirelessly from each other. Data from the sensors in the chair were sent to the Particle cloud, which Whiff was subscribed to. An ultrasonic scent diffuser was used to release the scent in Whiff. The scent was released by transposing high-frequency sound waves into mechanical energy that is transferred into a liquid, creating standing waves. As the liquid exits the atomizing surface of the disc, it's broken into a fine mist of uniform micron-sized droplets.

The housing and mechanisms of Whiff are all 3D printed with PLA. The 3D-printed housing is connected to a glass globe which made the smoke created by the ultrasonics scent diffuser visible to the user. The electronics of the system are placed on the bottom parts of Whiff. Scented oils were used in the diffuser to carry this scent via the fog, created with the diffuser. The fog which carried

the scent needed to be disposed based on the data. To control the release of the scent, a screw construction was 3D printed on top of the glass globe. The screw construction could be opened with a servo based on the data collected with the chair.

To summarize the goal and mechanism of Whiff: When an office worker is sitting behind their desk, the pressure sensor on the user's chair and flex sensor on the user's back measure that they have been sitting in the same position for too long (longer than 30 min). The bell jar slowly fills with fog to create awareness. Reaching a certain threshold, Whiff opens and releases a scent reminding individuals of the forest to raise awareness to go for a WorkWalk, switch from a sitting to a standing position, or take a break. The data physicalisation in Whiff is noticeable for both the individual and the social environment, targeting multiple people while creating awareness for a healthier office environment.

4.1.1 Space Mapping of Whiff. Whiff expresses nature-like-scents to create awareness and motivate individuals to have a more active workstyle. These scents, from a chemical feature (Figure 5), use nature-like chemicals such as Lyril (pine tree) and Lilial (Salty mariner) to create a nature-like scent. These nature-related scents are linked to enhancing the well-being of due to their relatedness towards plants, flowers, and water [60]. Therefore, these scents have an emotional feature where individuals can relate to the scent of being outside and be active.

From a spatial feature perspective, Whiff is placed near the desk of users to convey information about their posture and sedentary behavior. The scent is therefore oriented toward the user itself, however, the scent is also notifiable for the direct environment of the user. The movement of the scent will depend on the office context itself and the openness (closed office vs open office) of the building. From a temporal feature/perspective, Whiff releases a scent if the posture of a user is incorrect for a certain time. This is done in two phases; first Whiff diffuses fog in the glass globe. When the user stays seated the Whiff opens and the scent will be released.



Figure 4: Re-cafeen: an olfactory artifact that spreads the scent of coffee when the temperature in a meeting room becomes too high.

4.2 Re-cafeen

Re-cafeen (Figure 4) was developed to make individuals reflect on their meeting room environment. When the temperature in the meeting room rises, Re-cafeen will spread the scent of coffee. The temperature in the meeting room is measured with a temperature sensor (DHT22) and controlled with a Particle Photon. If the temperature rises above a critical point (e.g. 25 degrees Celcius) a heating pad in the device will turn on and heat the coffee grounds. Several experiments were conducted by the designers with (i) coffee oil, (ii) recycled coffee grounds, and (iii) coffee powder to learn which scent would have the most effect. Based on these experiments, coffee grounds were selected. During the heating process, the smell of coffee will spread over the meeting room, indicating to the users that it is too warm and leaving the meeting room for some fresh air and coffee. After the temperature in the meeting room lowers, the heating pad will turn off and the coffee scent will slowly disappear.

The electronics of Re-cafeen are placed in a bamboo box giving it an aesthetically pleasing feeling. On top of the artifact, there is a place for two test tubes that need to be filled with coffee grounds. The user can choose from several brands and flavors of coffee grounds based on their preferences.

To summarize the goal and mechanism of Re-cafeen: Coffee is one of the beverages most commonly drunk in the office environment. Therefore, the scent of coffee is often related to taking a break to get something to drink. Moreover, office workers often sit too long in a closed meeting room, resulting in a rise in temperature, a lack of fresh air, and a drop in concentration and productivity as a result of this. Re-cafeen is an artifact that spreads the scent of coffee as an indication that the temperature in a meeting room is becoming too high. When the temperature in a meeting room raises above a critical point, a heating element will turn on, heating the coffee and spreading the scent of coffee. The goal of the artifact is that individuals become aware of their surroundings and take a break from the unhealthy work environment.

4.2.1 Space mapping of Re-cafeen. Re-cafeen uses the scent of coffee as an olfactory data physicalisation module to make office workers reflect on their unhealthy meeting room. Several factors influence the aroma of coffee including chemical, biological, and physical influences of cultivar, coffee cherry maturity, geographical growing location, production, processing, and roasting [63]. When heating and/or roasting coffee results in the release and formation of numerous chemical compounds (Figure 5) breakdown of amino acids, and degradation of trigonelline, quinic acid, pigments, and lipids [10]. When looking at the emotional features of Re-cafeen, the coffee scent reminds office workers to take a break, due to the emotional link between coffee and breaks at work. After the scent is released, the process will be repeated after the users start working again.

The spatial feature/information of Re-cafeen is conveyed via the scent. The scent is spread within the meeting room where Re-cafeen is placed, which is usually in the center of the meeting room or table. The orientation and diffusion of the scent depend on the setting of the meeting (e.g., open space vs closed space) and openness of the meeting setting (scent being able to spread or not from a closed room due to open or closed door/window). For example, a meeting in a closed room could have a different/bigger effect due to a higher density of coffee, compared to the scent being spread in an open environment. The temporal feature of Re-cafeen is based on the meeting room. The scent of coffee will be spread for 5 minutes when the temperature in the meeting room is too high. After 5 min, the heating pad will turn off. If the temperature in the meeting room is still too high after the cooling down period, the heating pad will turn on again and the scent of coffee will be spread again.

5 DISCUSSION

In this paper, we explore and discuss the use of scent as a data physicalisation mechanism in the office setting. This was explored via two artifacts called Re-cafeen and Whiff. Both artifacts use different scent modalities based on different data types, and spatial placements to create healthier office environments. The artifacts



Figure 5: Smell space mapping of Re-cafeen and Whiff.

were evaluated using the Smell space mapping of Maggioni et al., [42]. Based on this evaluation, we reflect on the challenges for future work on this topic.

5.1 Person's relation toward a Scent

The scents used in the artifacts were evaluated by participants during the study to make sure the scent fit with the settings and the desired behavior. However, the experience of scents is very subjective and is different for individuals based on their preferences [19, 23, 47]. An example of this can be seen in Re-cafeen where the scent of coffee is used, while not all office workers might drink or like coffee. These user interfaces contain materials (in this case scent) that can carry embedded meanings for users from other experiences/contexts [17]. Therefore, due to the subjective relation of a person could have with a scent, the effect of the smell-based interface could lead to a different response than the intended behaviour. Office-related artifacts should therefore consider personal differences and preferences when using this sensory modality [42].

5.2 Desensitized toward the Scent

Both artifacts present a scent to create awareness of office workers and/or their work environment. The presented artifacts release their scent based on data (e.g., temperature or posture) for a set period. However, the released scent could stay in the environment for a longer time. This lingering effect of a scent, how long a scent stays in an environment, could therefore be longer than the intended feedback period [19]. Individuals might get desensitized to a certain scent to prevent an overload [45]. Frequent repetition and exposure to a pleasant smell can also decrease the pleasantness of the experience of that scent [19]. Additionally, individuals might not be able to notify scents (temporarily) due to, for example, having a cold. Combining an olfactory approach with, for example, visual, haptic, or auditorial feedback could address these potential issues.

We want to encourage designers to further explore this by taking a multi-sensory approach where different sensory stimuli are targeted within the development of future interventions to enhance the effect of the artifact in the office [17, 48, 50].

5.3 Situatedness of Scent in the Office

Olfactory artifacts present their data physicalisation within different office environments (e.g., meetings rooms, employee desks). Therefore, the situatedness of the data physicalisation in the environment plays a key role in the presentation of scents. This goes both ways, such as presenting data in a closed environment only visible to a select group for privacy reasons or presenting data in an open office environment to enhance the social effect of the data physicalisation [31, 40]. To enhance the social effect of olfactory artifacts, individuals should be placed in proximity of office workers to increase communication, potentially leading to an increase in cooperative activities [38, 59]. However, due to the unpredictable nature of spreading scent, future development of olfactory artifacts should consider the intended location of the artifact considering the audience and privacy before placing an olfactory artifact in context [4].

5.4 Limitations and Future research

The presented artifacts were tested for some initial user testing to evaluate their usability of the artifacts. However, no real long-term research on the artifacts was conducted to study the olfactory artifacts on the behavior of the office workers in context. To gain a better understanding of the effect of these artifacts on the behavior of office workers, a longitudinal study needs to be conducted within a larger work context. This study will have a more focused approach on the effect of the olfactory artifacts on a motivational and behavioral change level.

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